MIL-B-60410 (MU)
21 September 1966
SUPERSEDING
PA-PD-2657
26 February 1965

MILITARY SPECIFICATION

BLASTING MACHINE, HANDLE OPERATED, METAL CASE 10 CAP CAPACITY

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements for a blasting machine (hand operated generator) used for detonating electric blasting caps or electric squibs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood-cleated - Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner
PPP-B-636 - Box, Fiberboard
UU-T-111 - Tape: Paper, Gummed (Sealing and Securing)

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173 - Varnish, Moisture and Fungus Resistant, (for Treatment of Communications, Electronic and Associated Electrical Equipment)
MIL-T-704 - Treatment and Painting of Material

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

MIL-B-3106 - Board, Composition Water Resistant, Solid
MIL-L-10331 - Leather, Moisture and Mold Resistant Treatment
MIL-C-45662 - Calibration System Requirements
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PUBLICATIONS

ORDNANCE CORPS

ET-9200678 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 OTHER PUBLICATIONS

The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards:
Handbook H-28 - Screw Threads Standards for Federal Services.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Officer, Washington 25, D.C.).

American Standards Association Standards:
ASA B6.1 - 1932 - Spur Gear Tooth Form
ASA B6.6 - 1946 - Gear Tolerances and Inspection

(Applications for copies should be addressed to the American Standards Association, 10 East 40th street, New York 16, New York.):

National Electrical Manufacturers
Association Standards: NEMA Standards

(Applications for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York 17, New York).

Official Classification Committee Publication:
Uniform Freight Classification Rules.

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York).

3. REQUIREMENTS

3.1 Design.-The blasting machine shall be a manually handle operated, twisting type (clockwise) generator and shall be enclosed in a waterproof metal case. It shall be capable of providing ample current to detonate or fire 10 electric blasting caps or electric squibs properly connected in series. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair and replacement.

3.2 Materials.-Materials shall be as specified. Materials not specifically designated shall be of good commercial quality. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium shall be avoided wherever detrimental.

3.3 Construction.-The blasting machine shall be complete and conform to the applicable specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

3.3.1 Castings and forgings.-All castings shall be free from blow holes, porosity, hard spots, shrinkage defects, cracks or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching and other defects that may affect the strength of the forgings involved. Strength and other physical and chemical properties of the castings and forgings shall be adequate to meet the requirements specified.

3.3.2 Welding, brazing and soldering.-Welding, brazing, and soldering shall be performed in accordance with good commercial practice. In no event shall peening or plugging be used on castings or forgings for re-claiming any parts of the blasting apparatus without authorization of the procuring activity.

3.3.3 Fastening devices.-All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Standard parts.-MIL Standard parts shall be used where applicable. In event MIL-Standard parts are not available, commercial utility parts such as screws, nuts, bolts, etc. may be used.

3.3.5 Electrical connections.-Electrical connections shall be mechanically secured and soldered; except connections on screw type terminals which shall be mechanically secured.

3.3.6 Lubrication.-All bearings (except sealed for life type) and other working parts shall be accessible for lubrication. Apparatus shall be lubricated wherever necessary prior to delivery.

3.3.7 Safety devices.-Each blasting machine shall provide the operator with adequate protection against electric shock.

3.3.8 Interchangeability.-All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be replaced or adjusted without requiring modification.

3.3.9 Surfaces.-All castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, nicks, gouges, dents, and scratches exposing bare metal, flux and other harmful or extraneous materials. External surfaces shall be smoothed and all edges shall be rounded or beveled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

3.3.10 Protective finishes.-All exposed metal and working surfaces except electrical contact surfaces, shall be dull finished. Exposed metal is defined as metal exposed to atmospheric conditions.

3.3.11 Painting.-Unless otherwise specified by the procuring activity, surface preparation and paint application shall be in accordance with the requirements of Military Specification MIL-T-704, except that the blasting machine shall be finished with one coat of primer followed by two coats of enamel. (See 6.5 for color).

3.4 Performance and product characteristics

3.4.1 Life cycle.-The blasting machine shall be capable of firing standard electric blasting caps, or electric squibs with the current output specified in 3.4.2. The machine and handle shall be capable of withstanding a minimum of 5000 actuations without requiring component replacement or repair and without a decrease of current output below specified level and at a min. rate of 20 actuations per minute. Each actuation shall be at a rotational speed which results in the current output as specified in 3.4.2.

3.4.2 Capacity.-The blasting machine shall have a current output, sufficient to fire or detonate at least ten electric blasting caps or electric squibs properly connected in series, connected through a minimum of 500 feet of two conductor, insulated No. 18 (min.) copper firing cable. The machine shall deliver a minimum of 1.5 amperes DC through a minimum external resistance of 52 ohms for a duration of more than 10 milliseconds. In addition, it shall not be possible for the machine to deliver any current output that is less than 1.5 amperes, DC.

3.4.3 Characteristics, dimensions and weight.-The blasting machine shall be of sturdy construction, capable of withstanding rough handling and usage in the field. It shall not exceed a height of 6 1/4 inches, less handle and 8 1/2 inches with handle, width of 4 5/8 inches, depth of 3 5/8 inches. It shall not exceed a weight of six pounds.

3.4.4 Temperature and humidity conditions

3.4.4.1 Storage temperatures.-The blasting machine and all components shall be able to withstand storage without damage in a temperature of minus 65 degrees F. for a period of 72 hours, and a 24 hour cycle of temperature, starting with 10 hours at plus

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

90 degrees F., followed by 5 hours of steady increase to plus 155 degrees F., 4 hours at plus 155 degrees F., and 5 hours of steady decrease to plus 90 degrees F. Following these extreme exposures and when stabilized at plus 80 degrees F. plus or minus 10 degrees F., the blasting apparatus shall conform to the performance requirements of 3.4.1 and 3.4.2.

3.4.4.2 Operating temperature.-The blasting machine and all components shall withstand without damage and operate within the requirements set forth in 3.4.2, while subjected to operating temperatures of minus 40 degrees F. and plus 125 degrees F.

3.4.4.3 Humidity conditions.-The blasting machine shall operate within the requirements of 3.4.2, after exposure to 20 hours of relative humidity between 92 and 97 percent at air temperatures from plus 80 degrees F. to plus 85 degrees F., followed by 4 hours of 100 percent relative humidity with condensation at plus 75 degrees F. to plus 80 degrees F.

3.4.5 Vibration.-The blasting machine shall be capable of withstanding vibration of simple harmonic motion, for 1 hour in each of the three major axial directions, at an amplitude of 0.030 inch (0.060-inch total excursion) and at a frequency variation such that the entire range between 10 and 55 cycles per second is covered each minute. Following vibration, the blasting machine shall show no externally visible or audible evidence of damage and shall conform to the requirements of 3.4.1. and 3.4.2.

3.4.6 Shock.-The blasting machine shall withstand being dropped a vertical distance of 4 feet through each of its three major axes on a solid concrete surface. Following this test, the blasting machine shall conform to the requirements of 3.4.1 and 3.4.2.

3.4.7 Dielectric strength.-The blasting machine shall be capable of withstanding an application of a high potential applied for one minute between mutually insulated circuits and between insulated circuits and external metal parts. Potentials applied shall be 500 volts DC. Following this test, the blasting machine shall conform to the requirements of 3.4.1 and 3.4.2.

3.4.8 Waterproofness.-The blasting machine shall be capable of withstanding submersion under a four foot column of water, for a continuous period of at least eight hours. The temperature of the water to be approximately the same as that of the machine at the time of immersion. Following the submersion, the blasting machine shall show no evidence of water seepage and shall conform to the requirements of 3.4.1 and 3.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

3.5 Detail of components.-The blasting machine shall consist of the following principal component parts:

3.5.1 Generator.-The generator of the blasting machine shall be of the self-excited, shunt or compound wound direct-current type. The armature windings and field coils shall be carefully insulated and impregnated with suitable varnish or compound whose melting point shall be above 80 degrees C. Neither shellac nor paraffin shall be used for impregnating windings or coils. The material and construction of the magnetic circuit shall be such as to retain a strong residual magnetism.

3.5.2 Operating mechanism.-The operating spindle shall extend through the top of the enclosure and shall be operated by a detachable handle, which shall be secured by a chain to the blasting machine. The protruding portion of the operating spindle shall be enclosed within a shoulder or collar so designed that the operating spindle cannot be operated except by the handle. The operating handle shall be proportioned and dimensioned with at least 1 1/2 inches minimum clearance between the under side of the hand grip of handle and the top of any machine obstruction, to permit the operator a firm grasp without striking the knuckles, during the twisting or turning movement. Proper function shall be accomplished with one twist of the handle a maximum of 150 degrees. The operating spindle shall cause the armature of the generator to revolve at an adequate speed and for the proper time duration for the detonation of electric blasting caps as specified in 3.4.2. White metal shall not be used for gears, pinions or any moving parts.

3.5.3 Enclosure.-The enclosure shall be a rugged, durable, waterproof metal case.

3.5.3.1 Carrying strap.-A leather carrying strap not less than 3/4 inch wide and not less than 5/64 inch thick shall be fastened securely to the body of the blasting machine in such a manner that it can also be used as a stirrup on the under side of the machine, through which the operator may slip his hand for assistance in holding the machine firmly in one hand while operating the handle with the other.

3.5.3.2 Binding posts.-Two binding posts, with suitable knurled thumb nuts which cannot be removed, shall be mounted on the top of the case, so proportioned as to permit wires as small as 0.032 inch in diameter and as large as 0.081 inch in diameter to be firmly clamped by the thumb nuts. Binding posts and thumb nuts shall be nonferrous, conductive metal. Thumb nuts shall be finished in accordance with paragraph 3.3.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

3.6 Electrical components.-Electrical components of the blasting machine, including wiring and their enclosures shall conform to and shall be located in accordance with the NEMA Electrical Standards, except as specified by the procuring activity. No component parts shall be used in any application where the requirements exceed the manufacturer's rating for the part.

3.7 Fungus treatment.-The blasting machine shall be treated for fungus resistance in accordance with the requirements of Military Specification MIL-T-152, using material conforming to requirements of Military Specification MIL-V-173. Leathers shall be treated for moisture and mold resistance in accordance with the requirements of Military Specification MIL-L-10331.

3.8 Identification of product.-The blasting machine shall be marked for identification in accordance with Military Standard MIL-STD-130.

3.9 Instruction plates.-A brass instruction plate shall be attached to each blasting machine giving Federal stock number and manufacturer's serial number, the rated capacity expressed in the number of electric blasting caps, each having copper leg wires 30 feet in length, that can be fired in series, and the proper method of operating the unit. The instruction plate shall include the notation: "This machine may lose capacity temporarily; to restore, twist handle vigorously a few times before connecting circuit."

3.10 Handbook of instructions.-The contractor shall furnish one copy of a handbook of instructions with each machine supplied under this specification. The handbook shall contain necessary operating instructions, drawings, testing instructions, illustrations, parts lists, manufacturer's part numbers, lubrication instructions, assembly and disassembly instructions and safety precautions, to insure proper operation and maintenance of the blasting machine. A preliminary copy of the instruction handbook shall be submitted to the contracting officer for approval prior to publication and delivery.

3.11 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. The blasting machine shall be free from any defects which may affect serviceability and conformance with good commercial practices.

3.12 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time of the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of five (5) assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials, (see 4.1.1e). A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
- Check the source to verify that this is the current version before use.

4.3.2 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Construction (see 3.3)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 1.5 percent		
101.	Overall height (with and without handle), max. (see 3.4.3)....	Gage	01001
102.	Overall width, max (see 3.4.3)...	Gage	01002
103.	Overall depth, max. (see 3.4.3)..	Gage	01003
104.	Weight, max. (see 3.4.3).....	Scale	01004
105.	Safety devices, not as specified (see 3.3.7).....	Visual-Manual	01005
106.	Interchangeability, not as specified (see 3.3.8).....	Visual-Manual	01006
107.	Generator, not as specified (see 3.5.1).....	Visual-Manual	01007
108.	Any defect affecting serviceability (see 3.5.2).....	Visual-Manual	01008
109.	Handbook or instructions, not as specified (see 3.10).....	Visual	01009
110.	Brass instruction plate, not as specified (see 3.9).....	Visual	01010
111.	Electrical connections, not as specified (see 3.3.5 and 3.6)....	Visual-Manual	01011

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the currentversion before use.

112.	Electrical equipment, not as specified (see 3.3.5 and 3.6)....	Visual	01012
113.	Component missing or not as specified (see 3.5).....	Visual-Manual	01013
114.	Terminals, deformed.....	Visual	01014
115.	Leads damaged or improper insulation.....	Visual	01015
Minor:	AQL 4.00 percent		
201.	Dissimilar metals, contacting....	Visual	01016
202.	Peening or plugging.....	Visual	01017
203.	Fastening devices not tight or improper.....	Visual-Manual	01018
204.	Evidence of flux.....	Visual	01019
205.	Lubrication, not as specified (see 3.3.6).....	Visual	01020
206.	Marking missing or improper.....	Visual	01021
207.	Protective finish omitted or improper.....	Visual	01022
208.	Color, improper.....	Visual	01023
209.	Evidence of poor workmanship (see 3.11).....	Visual	01024

4.3.2.2 Preservation and packaging (see 5.1).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Preservation, not as specified...	Visual	02001
102.	Unit packaging, not as specified.	Visual	02002
103.	Packing, not as specified.....	Visual	02003
104.	Marking, not as specified.....	Visual	02004
Minor:	None defined		

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

4.3.3 Testing

4.3.3.1 First article sample (see 4.2.1).-The entire sample submitted in 4.2.1 shall be tested as specified after being examined for all dimensions, specification and contractual requirements. Rejection and resubmission shall be as specified in 4.2.3. The sequence of testing shall be as follows:

		Code No.
4.3.3.1.1	Capacity (see 4.4.2)	03001
4.3.3.1.2	Shock (see 4.4.6)	03002
4.3.3.1.3	Vibration (see 4.4.5)	03003
4.3.3.1.4	Operating temperature (see 4.4.4)	03004
4.3.3.1.5	Humidity conditions (see 4.4.3.3)	03005
4.3.3.1.6	Storage (see 4.4.3.1 and 4.4.3.2)	03006
4.3.3.1.7	Waterproofness (see 4.4.8)	03007
4.3.3.1.8	Dielectric strength (see 3.4.7 and 4.4.7)	03008
4.3.3.1.9	Life cycle (see 3.4.1 and 4.4.1)	03009
4.3.3.1.10	Waterproofness (enclosure opened) (see 3.4.8 and 4.4.8)	03010
4.3.3.1.11	Dielectric strength (see 3.4.7 and 4.4.7) (enclosure opened)	03011

4.3.3.2 Production testing

4.3.3.2.1 Production Acceptance Tests.-The sample size shall be selected in accordance with Table 1, special inspection level S4 and Table 2A, of Military Standard, MIL-STD-105D. The sample shall be tested for the following requirements in the sequence indicated:

Test Description	Test Paragraph	Category	Code No.
Capacity	4.4.2	Major	03001
Operating Temp.	4.4.4	Major	03004

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

Thereafter only 25 percent of sample size shall be tested for the following requirements in the sequence indicated:

Test Description	Test Paragraph	Category	Code No.
Water proofness	4.4.8	Major	03007
Dielectric Strength	4.4.7	Major	03008
Life Cycle	4.4.1	Major	03009
Water proofness (Enclosure opened)	4.4.8	Major	03010
Dielectric Strength (Enclosure opened)	4.4.7	Major	03011

Failure of one or more units of the sample in meeting any of the requirements shall reject the lot. The Government, in addition, reserves the right to perform any or all of the tests as specified in this specification (see 4.3.3.1) at unscheduled or scheduled intervals or at the time of any or all production lots submitted for Government acceptance, where in the discretion of the Government, such tests are deemed necessary to assure full conformance with the prescribed requirement of the contractual documents.

4.3.3.2.2 Capacity (see 3.4.2) - Major defect - Code No. 03001.-All samples submitted shall be subjected to this test. The test shall be performed as specified in 4.4.2 using test equipment specified in 4.3.4.

4.3.3.2.3 Shock (see 3.4.6) - Major defect - Code No. 03002.- The test shall be performed as specified in 4.4.6 using test equipment specified in 4.3.4.

4.3.3.2.4 Vibration (see 3.4.5) - Major defect - Code No. 03003.-The test shall be performed as specified in 4.4.5 using test equipment specified in 4.3.4.

4.3.3.2.5 Operating temperature (see 3.4.4.2) - Major defect - Code No. 03004.-All samples submitted shall be subjected to this test. The test shall be performed as specified in 4.4.4 using test equipment specified in 4.3.4.

4.3.3.2.6 Humidity condition (see 3.4.4.3) - Major defect - Code No. 03005.-The test shall be performed as specified in 4.4.3.3 using test equipment specified in 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the currentversion before use.

4.3.3.2.7 Storage (see 3.4.4.1) - Major defect - Code No. 03006.-The test shall be performed as specified in 4.4.3 using test equipment specified in 4.3.4.

4.3.3.2.8 Waterproofness (see 3.4.8) - Major Defect - Code No. 03007.-The test shall be performed as specified in 4.4.8 using test equipment specified in 4.3.4.

4.3.3.2.9 Life cycle (see 3.4.1) - Major defect - Code No. 03009.-The test shall be performed as specified in 4.4.1 using test equipment specified in 4.3.4.

4.3.3.2.10 Dielectric strength (see 3.4.7) - Major Defect - Code No. 03008.-The test shall be performed as specified in 4.4.7 using test equipment specified in 4.3.4.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-9200678 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.4.1 Life cycle test.-The blasting machine with handle attached shall be operated for a total of 5000 actuations at a minimum rate of 20 actuations per minute. Each actuation shall be at a rotational speed which results in the current output, as specified in 3.4.2. During actuations a minimum of six photographs shall be taken of the current output as displayed on the ordinate of an oscilloscope grid (see 4.4.2). The photographs shall be taken at the approximate actuations of 1, 1000, 2000, 3000, 4000, and 5000. Any evidence of wear damage or failure to meet the requirements of 3.4.1 and 3.4.2 shall constitute failure of the test. Upon completion of this test, the blasting machine shall be disassembled to allow for examination of all its assemblies or components. The dielectric strength requirement (see 3.4.7) and the waterproofness requirement (see 3.4.8) shall be tested or verified on all assemblies, or components, which could not be tested or verified prior to disassembly of the machine (see 4.4.7 and 4.4.8). Any evidence of wear, damage, dielectric failure or water seepage, shall constitute a failure of the test.

4.4.2 Capacity (current output) test.-The blasting machine shall be tested to determine its capability of meeting the requirements of 3.4.2. Test shall be performed with an oscilloscope displaying clearly readable specified time and currents on the ordinates of its CRT grid. The oscilloscope shall have a photographic attachment. The minimum 1.5 amperes value shall be the sum of the current from zero to the average minimum peaks plus seventy percent of the current between this value and the average maximum peaks. Failure to deliver the minimum current for the minimum time shall be cause for rejection of the lbasting machine (see 3.4.2.)

4.4.3 Storage tests

4.4.3.1 Low temperature test.-The blasting machine shall be placed in an air temperature of minus 60 degrees F. to minus 65 degrees F. for a period of 72 hours. At the end of this time, the blasting machine shall be removed and allowed to stabilize at an air temperature of plus 80 degrees F. plus or minus 10 degrees F. The blasting machine shall then be inspected for evidence of damage and tested for conformity to 4.4.2 (see 3.4.4.1).

4.4.3.2 High temperature test.-The blasting machine shall be subjected to a 24 hour cycle of temperature starting with 10 hours

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

at plus 90 degrees F. plus or minus 5 degrees F., followed by 5 hours of steady increase to plus 150 degrees F. to plus 155 degrees F., 4 hours at plus 150 degrees F. to plus 155 degrees F., and 5 hours of steady decrease to plus 90 degrees F., plus or minus 5 degrees F. The blasting machine shall then be inspected for evidence of damage and tested for conformity with 4.4.2 (see 3.4.4.1).

4.4.3.3 Humidity test.-The blasting machine shall be subjected to 20 hours exposure with relative humidity between 92 and 97 percent and at air temperature from plus 80 degrees F. to plus 85 degrees F. followed by 4 hours of 100 percent relative humidity with condensation at plus 75 degrees F. to plus 80 degrees F. At the conclusion of the humidity test the blasting machine shall be inspected for evidence of damage and tested for conformity with 4.4.2 (see 3.4.4.3).

4.4.4 Operating temperature test

4.4.4.1 Low ambient temperature test.-The blasting machine shall be subjected to an air temperature of minus 35 degrees F. to minus 40 degrees F. until stabilized at 40 degrees F. While at that temperature, it shall be tested for capacity in accordance with 4.4.2 (see 3.4.4.2).

4.4.4.2 High ambient air temperature test.-The blasting machine shall be subjected to an air temperature of plus 120 degrees F. to plus 125 degrees F. until stablized at plus 125 degrees F. While at that temperature it shall be tested for capacity in accordance with 4.4.2 (see 3.4.4.2).

4.4.5 Vibration test.-The blasting machine shall be subjected to a continuous vibration of simple harmonic motion for 1 hour in each of the three major axial directions at an amplitude of 0.030 inch (0.060-inch total excursion) and at a frequency variation such that the entire range between 10 and 55 cycles per second is covered each minute. At the conclusion of the test, the blasting machine shall be inspected for evidence of damage and tested for conformity with the requirements of 4.4.2. Evidence of damage or failure to perform satisfactorily shall be cause for rejection of the blasting machine (see 3.4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

4.4.6 Shock test.-The blasting machine shall be dropped a total of three times, through each of its major axes, a vertical distance of four feet onto a solid concrete surface. One of said drops shall be with the spindle facing down. Following this test, the machine shall show no evidence of damage causing failure to conform to the requirements of 3.4.2. Evidence of failure to perform satisfactorily shall be cause for rejection (see 3.4.6).

4.4.7 Dielectric strength test.-Prior to life cycle test and disassembly, the blasting machine shall be subjected to 500 volts DC applied for one minute between mutually insulated circuits and between insulated circuits and external metal parts. The machine shall meet the requirements of 3.4.1 and 3.4.2 after which the machine shall be disassembled. The circuits and assemblies which were not previously tested because of inaccessibility shall be subjected to this test. The machine shall be reassembled and shall conform to the requirements of 3.4.2. Evidence of damage or dielectric failure will be cause for rejection of the machine (see 3.4.7).

4.4.8 Waterproofness test.-The blasting machine shall be submerged for a minimum of 8 hours, in a water tight container, under a four foot column of water (minimum). The water shall be mixed thoroughly with fluoroscene color concentrate at a ratio of one ounce of fluoroscene color concentrate to each one gallon of water in the container. The temperature of the water to be approximately the same, as that of the machine, at the time of immersion. Following the submersion, the machine shall conform to the requirements of 3.4.2. After the machine has been subjected to the life cycle test, it shall be disassembled and examined for evidence of water seepage and examined under a black light for internal evidence of fluoroscene material. Evidence of water seepage or fluoroscene shall be cause for rejection (see 3.4.8).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-Preservation and packaging shall be level A or C as specified in the contractual documents and applicable specifications.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the currentversion before use.

202. Metal ends loose or distorted

5.1.1 Cleaning.-Each blasting machine shall be cleaned method C-1 in accordance with the requirements of Military Specification MIL-P-116.

5.1.2 Drying.-The blasting machine shall be dried in accordance with one or more of procedures D-1 through D-4 of Military Specification MIL-P-116.

5.1.3 Unit Packaging.-Each blasting machine, comprising a complete unit, shall be individually preserved and packaged in accordance with method 1A-15 (Carton Barrier Bag) in accordance with the requirements of Specification MIL-P-116. Sealed bags shall meet the heat seal requirements of MIL-P-116.

5.2 Packing

5.2.1 Level A.-The blasting machine shall be packed in a nailed wooden box conforming to PPP-B-621, Class 2, Style 2, 2 1/2 or 4 or PPP-B-601 overseas type, Style A or B. The sides and ends of the box shall be lined with Composition Board, MIL-B-3106. Additional composition board shall be added to the top and sides of the box as required for a tight pack. The gross weight of the box shall not exceed 200 pounds. Boxes shall be closed and strapped in accordance with appendix of the box specification.

5.2.2 Level C.-The blasting machine shall be packed same as level A except boxes shall be PPP-B-621, Class 1, PPP-B-601 domestic type, or PPP-B-636, Type CF, Class Domestic variety SW, Grade 275. Gross weight of fiber boxes shall not exceed 70 pounds. Sides, ends and tops of boxes shall be lined with commercial paperboard or double faced corrugated board when level A packaged are packed in wooden boxes.

5.3 Marking.-In addition to any special markings required by the contractor or order, each unit interior and exterior shipping containers shall be marked in accordance with the requirements of Military Standard MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the EL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND5
SMUPA-DC8

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

6.5 Color.-The color shall be specified by the contracting officer.

6.6 Definitions

6.6.1 Electric blasting cap or squib.-A device which transforms electrical energy into heat which in turn activates an explosive force strong enough to initiate the detonation of a charge of cap or squib sensistive explosives.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1375-A1042

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:20Z
Check the source to verify that this is the current version before use.